

ADELAIDE UNIVERSITY

SCHOOL OF BIOLOGICAL SCIENCES

Before you Re-Sequence: Resolution of
Tandemly Repeated Vomeronasal Genes
using Existing Data Overturns Systematic
Underestimates in Sea Snakes (*Hydrophis*)

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Literature Review

Rapid Adaptive Radiations & Secondary Marine Transitions

Secondary Marine Adaptations as a Model for Rapid Phenotypic Change

The adaptation of species to their environment due to natural selection defines a major theme in evolution. When organismal lineages colonise significantly different environments, the selective landscape shifts across phenotypic systems simultaneously: locomotion, respiration and sensory detection are all placed under renewed pressure (Sanders et al., 2010). This is heightened in adaptive radiations, a specific case of diversification. This phenomenon is defined by the quick divergence of lineages into ecologically distinct forms following access to new niche opportunities. This may derive from simple processes, such as evasion of predation or competition. Alternatively, radiations can be catalysed by the acquisition of adaptive traits, known as key innovations, that enable the exploitation of vacant niches, or colonisation of new habitats (Miller et al., 2023).

Secondary marine transitions may present as one of the most extreme examples of environmental change, and have occurred at least 7 times independently in amniotes (Formoso et al., 2023). Convergent marine-associated phenotypes are pervasive amongst these lineages. Dorsoventral body compression, paddle-shaped appendages, modified respiratory systems, and convergent DNA amino acid substitutions have all been documented and confirm the challenges of natural selection (Formoso et al., 2023; Guo et al., 2024; Ludington et al., 2023; Peng et al., 2020).

From Land to Sea: Radiation of Hydrophiinae Snakes

The macroevolutionary transition from land to sea in Hydrophiinae (family Elapidae) has produced a striking concentration of phenotypic innovation within a single subfamily, evident in them exhibiting the greatest phylogenetic resolution and success among squamate marine transitions (Li et al., 2021; Sanders et al., 2010). Having derived from both a secondary marine transition and rapid radiation, *Hydrophis* sea snakes serve as a tractable model for studying such systems.

Diverging from Australo-Papuan terrestrial elapids within the last 9–18 million years (Ludington et al., 2023; Sanders et al., 2008), and speciating, almost simultaneously 1Ma, the ~48 species of *Hydrophis* collectively exhibit the full suite of convergent traits that characterise secondary marine vertebrates more broadly (Kishida et al., 2025; Peng et al., 2020). Dorsoventrally compressed bodies and paddle-shaped tails generate propulsive thrust (Brischoux & Shine, 2011; Sanders et al., 2012), whilst specialised respiratory traits, including haemostatic nostrils, extensive cutaneous gas exchange, and a redistributed cephalic vascular plexus, support prolonged submergence underwater (Heatwole & Seymour, 1975). These morphological and physiological shifts are underpinned by genome-wide signatures of positive selection enriched for hypoxia tolerance and circulatory development (Ludington et al., 2023). Population analysis of 14 *Hydrophis* species recovered strict reproductive isolation even between geographically distinct conspecifics (Kishida et al., 2025). By contrast, *Laticauda* (amphibious sea kraits) and *Aipysurus-Emydocephalus* (fully marine sea snakes) comprise 8 and 10 species, respectively, and show no comparable speciation acceleration (Kishida et al., 2025). However, the rationale for this delayed radiation, rather than gradual accumulation, remains unknown. The rapid radiation of *Hydrophis* has been linked to ecological specialisation which likely imposed strong divergent selection and accelerated lineage diversification (Sherrat et al., 2018). Vomeronasal-mediated mate-recognition systems likely mediated this radiation, as shifts in chemosensory signaling are predicted to both facilitate divergence and maintain species boundaries in sympatric lineages (Kishida et al., 2019). In a rapid radiation of a marine transition, selective

pressures are compounded for traits suited to the new shift in dominant signal medium (air to water).

Sensory Adaptation as a Hallmark of Aquatic Transition

The phenotypic profile of the land to sea transition is broad. Genomics has exposed hypoxia adaptation, immune response, sensory perception, alongside gene families associated with marine function – ion transport, skin permeability and chemoreception – all as being positively selected for, relative to terrestrial counterparts (Ludington et al., 2023). Among these dimensions, sensory system reorganisation is acute. Each new environmental and ecological niche imposes distinct demands on the detection and processing of environmental signals. Hence, the genes that encode these traits under selection represent accessible signatures to adaptation.

The visual system of Hydrophiinae provides parallel evidence. In teleost fishes, opsin gene repertoires track photic niche (spectral matching to the most abundant wavelength to maximise photon capture; UV is filtered with depth) across approximately 30,000 species, with deep-sea lineages contracting short-wavelength receptor classes while shallow-reef cichlids independently expand spectrally diverse variants (Musilova et al., 2019). Within Hydrophiinae, the visual system shows parallel modality-specific remodelling: Simões et al. (2020) reported trans-specific polymorphisms in the short-wavelength-sensitive opsin gene *SWS1* and interpreted the pattern as evidence for long-term balancing selection at a single locus. Subsequent analysis by Rossetto et al. (2024) revealed repeated *SWS1* duplications in multiple *Hydrophis* species, demonstrating that the same variation is produced by multiple *SWS1* gene copies, each with a fixed spectral type, rather than by allelic polymorphism at a single locus, contrasting sharply with opsin pseudogenisation in cetaceans and several other secondarily marine tetrapods (Rossetto et al., 2023). These duplicated opsins are simultaneously expressed in retinal tissue and together expand the snake's capacity for spectral sensitivity through greater copy number variation (CNV).

These findings confirm that the *Hydrophis* radiation is capable of selective genomic remodelling in direct response to the physical properties of the marine environment. The shift from air to water as the

dominant signal medium is equally dramatic for chemical detection, as airborne olfactory cues, the primary chemical signal on land, are absent underwater, and prey detection, mate recognition, and navigation necessarily shift toward reliance on dissolved compounds (Mollo et al., 2014). The vomeronasal system, which is structurally pre-adapted for precisely this class of signal, is therefore an ideal candidate system for studying the genomic consequences of the marine transition.

Chemosensory Biology

The Vomeronasal Organ: Structure and Function

Snake chemosensation comprises two parallel pathways that differ in anatomy, receptor chemistry, and ecological function. The main olfactory system (MOS) detects volatile, airborne compounds through olfactory receptor (OR) neurons in the main olfactory epithelium within the nostrils (Han et al., 2023). The vomeronasal system (VNS) operates in anatomical and functional parallel: specialised sensory neurons in the vomeronasal organ (VNO) located in the roof of the mouth detect dissolved, non-volatile molecules and project to the accessory olfactory bulb (Francia et al., 2014; Wong et al., 2018). Marine hydrophiines are the only vertebrates with a functional VNS and highly regressed MOS (Kishida et al., 2019). In squamates, and in snakes particularly, the VNS is the dominant chemical sense: the VNO contains more sensory neurons than the main olfactory epithelium, and lesion experiments have established that VNS-mediated input, not MOS input, is required for prey trailing, conspecific recognition, and mate localisation (Halpern et al., 1983). The snake tongue-flicking mechanism is uniquely pre-adapted for these cues: unlike the mammalian VNO, which requires negative-pressure nasal airflow for chemical delivery and fails underwater, the snake system transfers tongue-borne chemical cues into specialised vomeronasal ducts, where they are transported in a fluid phase to the sensory epithelium, rendering the system fully functional when submerged, in both

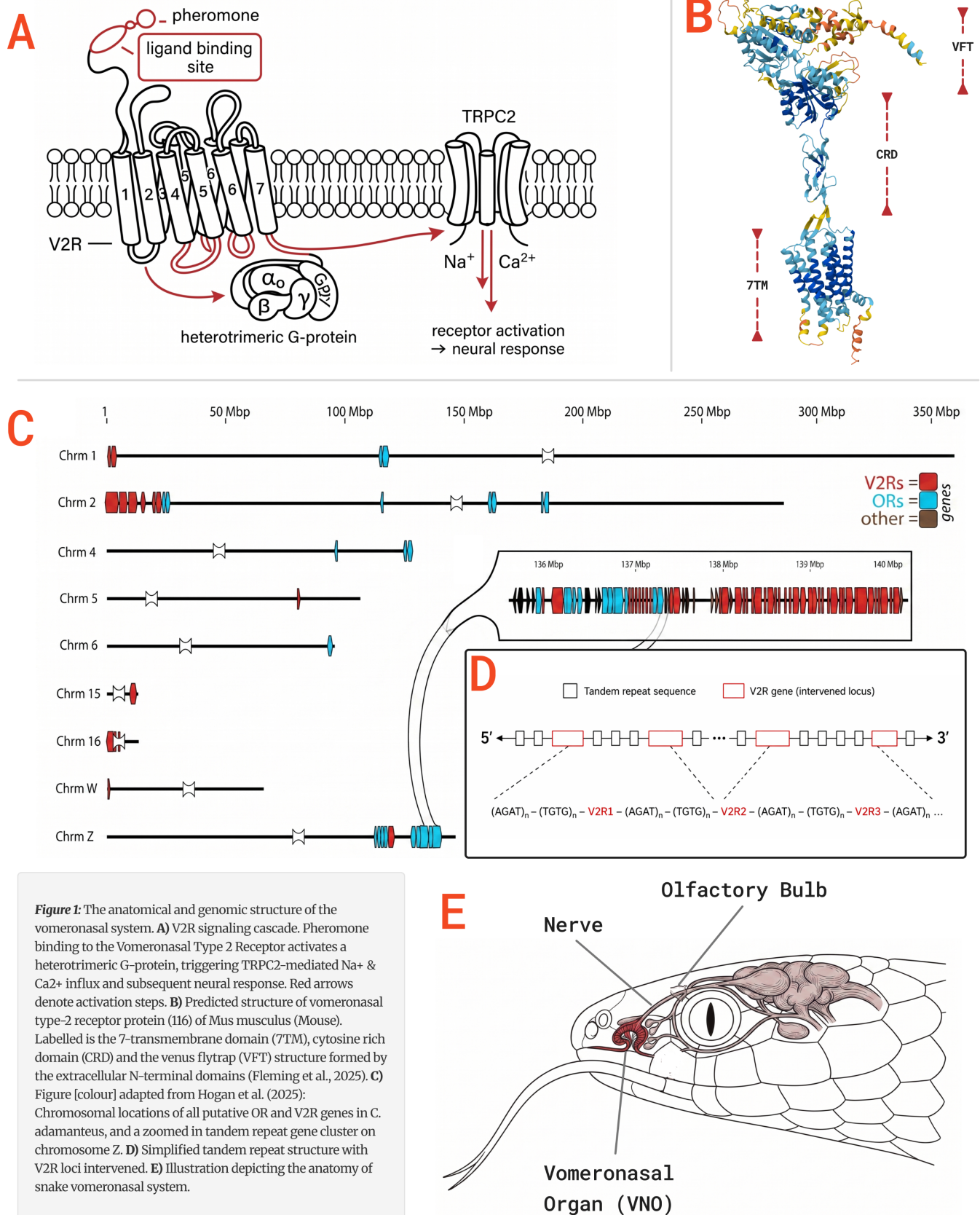


Figure 1: The anatomical and genomic structure of the vomeronasal system. **A)** V2R signaling cascade. Pheromone binding to the Vomeronasal Type 2 Receptor activates a heterotrimeric G-protein, triggering TRPC2-mediated Na⁺ & Ca²⁺ influx and subsequent neural response. Red arrows denote activation steps. **B)** Predicted structure of vomeronasal type-2 receptor protein (116) of *Mus musculus* (Mouse). Labelled is the 7-transmembrane domain (7TM), cytosine rich domain (CRD) and the venus flytrap (VFT) structure formed by the extracellular N-terminal domains (Fleming et al., 2025). **C)** Figure [colour] adapted from Hogan et al. (2025): Chromosomal locations of all putative OR and V2R genes in *C. adamanteus*, and a zoomed in tandem repeat gene cluster on chromosome 2. **D)** Simplified tandem repeat structure with V2R loci intervened. **E)** Illustration depicting the anatomy of snake vomeronasal system.

terrestrial and marine species, and hence explains the divergence in vomeronasal direction between marine amniotes (e.g., loss of VNO in cetaceans and sirenians).

Chemosensation in Marine vs. Terrestrial Taxa, and Evidence for Adaptation

The two chemosensory systems express distinct receptor gene families whose structural differences encode their respective ligand chemistry. Olfactory receptor neurons express class A GPCRs tuned to volatile, hydrophobic small molecules. Vomeronasal neurons express either vomeronasal type-1 receptors (V1Rs, class A) or vomeronasal type-2 receptors (V2Rs, GPCRs) (Brykczynska et al., 2013). In snakes, there are tendencies for the V1R complement to contract, whereas V2R diversity has *at least* been maintained consistently, with *Crotalus adamanteus* (eastern diamondback rattlesnake) exhibiting expression of >116 distinct V2R transcripts (minimum estimate) in VNO tissue, representing one of the highest V2R diversities recorded in vertebrates (Brykczynska et al., 2013).

Individual vomeronasal sensory neurons co-express one broadly distributed V2R-C family member with one lineage-specific non-C family receptor (Brykczynska et al., 2013). This bigenic pattern is conserved across vertebrates, reflecting an ancestral, functional requirement for receptor structure in signal transduction (Brykczynska et al., 2013). Understanding of this gene family characterises the dominant sensory channel of *Hydrophis*. The sensory phenotype of the VNS and its adaptive trajectory are thus encoded directly in the composition and diversity of V2R genes.

The genomic evidence across Hydrophiinae makes this contrast between OR and VNO quantitative. Fully marine hydrophiines carry a mean of 63.5 ± 14.9 intact OR loci, a significant reduction from their terrestrial elapid relatives, which average 335 ± 40.8 , while amphibious *Laticauda* retains an intermediate 114 ± 12.7 , possibly denoting grading with degree of marine dependence (Kishida et al., 2019). The volatile-amine receptor gene *TAAR2*, intact in terrestrial relatives, is pseudogenised in *Hydrophis*, reflecting directional loss of receptor capacity for volatile-phase ligands (Kishida et al., 2019).

Expression profiling confirms that V2Rs are actively transcribed in VNO tissue whilst OR expression is largely absent from the nasal epithelium, whilst also greater in juvenile individuals (Hogan et al., 2025; Peng et al., 2020). This suggests a degree of coordination between sensory programming of the VNO and development, in addition to validating that gene-count differences reflect functional investment rather than neutral variation.

The Vomeronasal Type-2 Gene Family: Biology, Dynamics, and Ecological Significance

V2Rs are class C GPCRs, structurally distinct from both olfactory receptors and V1Rs (class A). Their canonical architecture places the orthosteric (primary) ligand-binding site within a large extracellular ‘venus flytrap’ module (VFT; see figure 1), connected via a nine-cysteine linker domain to seven transmembrane helices (Brykczynska et al., 2013; Policarpo et al., 2024). Ligand binding triggers a calcium-signalling cascade that gates the ion channel responsible for converting receptor activation into a neural response. This signalling pathway is anatomically compartmentalised. V2R neurons occupy the basal region of the VNO, segregated from parallel chemosensory streams within the same organ, indicating functional partitioning of chemosensory streams within the organ itself (Francia et al., 2014). The functional consequence of this structure is extended ligand scope: the solvent-exposed binding cleft accommodates polar, high-molecular-weight compounds that are structurally incompatible with V1Rs (Brykczynska et al., 2013; Policarpo et al., 2024). Therefore, expansive diversification of the gene family may have facilitated greater chemical discrimination for sea snakes, enhancing navigation, predation and social functions.

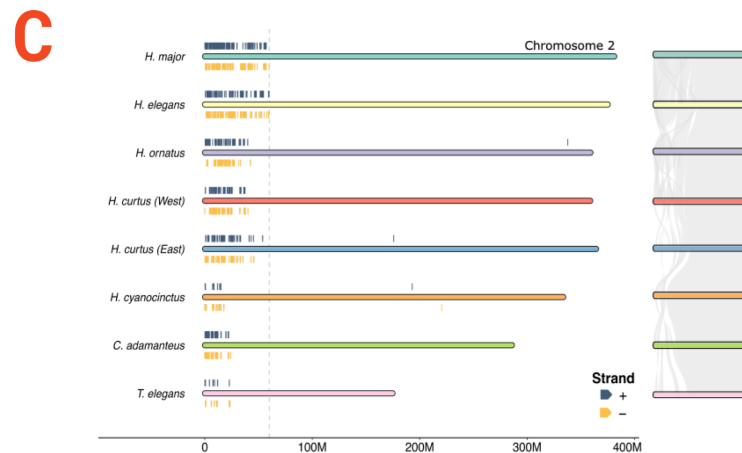
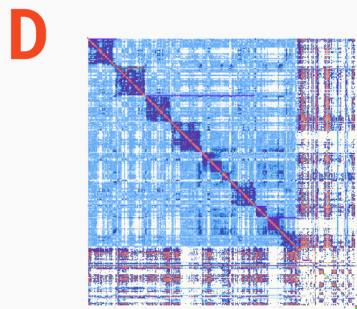
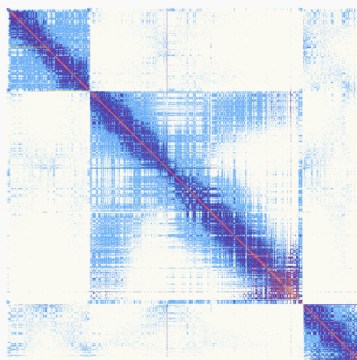
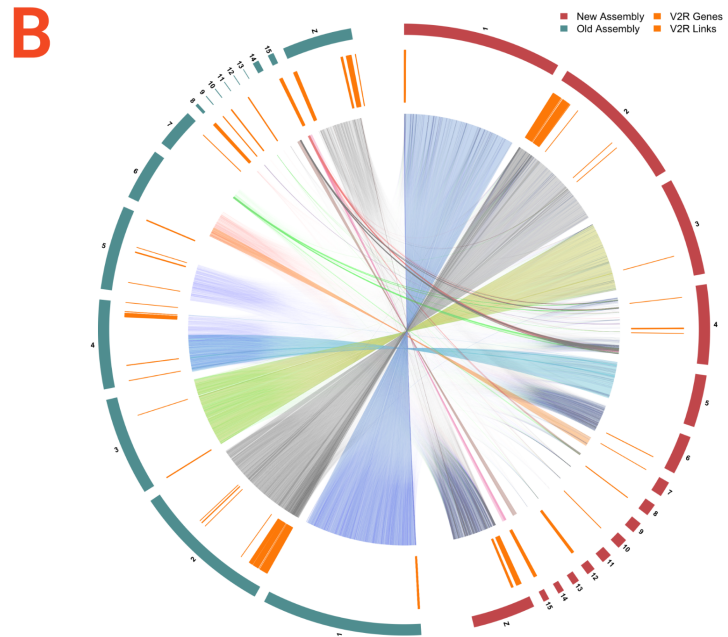
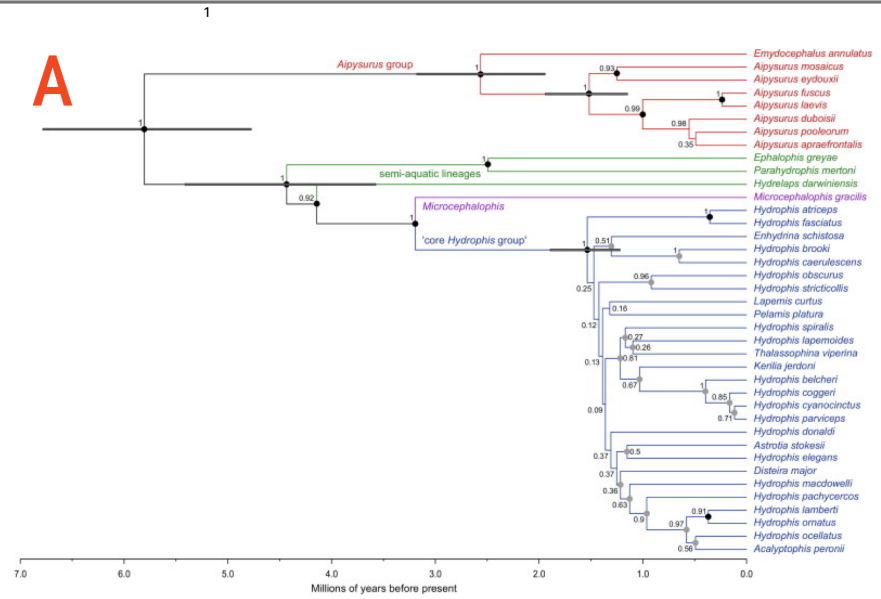
Gene family dynamics

In the genome, V2R genes are consistently organised into dense tandem arrays, in which multiple receptor genes are arranged within a single chromosomal region, often spanning several megabases and

rich in repetitive elements. Vertebrate V2R arrays evolve primarily through tandem duplications, generating paralogues that are either retained and functionally diverge or accumulate inactivating mutations and are eventually deleted (Alioto & Ngai, 2005; Young & Trask, 2007; Yang et al., 2005). This process, unlike concerted evolution (multigene families evolving 'in concert') through gene conversion (non-reciprocal transfer from homologous sequence, hence 'updating' each other), is capable of generating genuinely novel receptor variants over evolutionary time (Nei & Rooney, 2005). Two layers determine V2Rs' capacity for chemical discrimination, the repertoire size and the co-expression of extracellular receptor subunits per neuron (Silvotti et al., 2007). As each receptor encodes a distinct ligand profile, a larger repertoire covers proportionally greater chemical space, whilst the bigenic co-expression produces complexes whose combined chemical specificity exceeds that of either subunit alone (Brykczynska et al., 2013; Nei & Rooney, 2005). As such, the V2R gene family is not well described as a simple receptor catalogue because the relevant phenotype does not relate to a single receptor function, but rather ensembles the discrimination capacity: the range of dissolved chemicals resolvable by the repertoire acting collectively. Hogan et al. (2025) validated this basis in their findings of greater proportions of sites under selection in extracellular regions, compared to intracellular, consistent with these regions performing conserved functions. Essentially, the substrate of vomeronasal function does not rely on a single gene, instead on an extended, repetitive chromosomal region. Evolution and origin of vomeronasal-type odorant receptor gene repertoire in fishes

Mean V2R counts range from zero in cetaceans and sirenians — consistent with their independent VNO losses following their marine transitions — to 140.6 in amphibians, with *Xenopus tropicalis* encoding 578 complete V2Rs, implying a strong positive association between dissolved-compound detection and V2R gene diversity across vertebrates (Policarpo et al., 2024). In teleost fishes, V2R homologues (OlfC genes) form clusters of up to 88 genes across 16 subfamilies, confined to one or two chromosomal regions per species, indicating that chromosomal clustering of V2Rs predates the evolution of the tetrapod vomeronasal organ (Hashiguchi & Nishida, 2006). In monitor lizards (Varanidae), V2Rs are localised to chromosome Ch2 ends and to the ChZ/W sex chromosomes, with elevated copy number relative to non-varanid lizards (Matsubara et al., 2014; Zhu et al., 2026). Among

Figure 2. A) Maximum credibility species tree inferred by *BEAST analysis of 2–6 individuals per 39 species for six loci. Node labels show posterior probabilities. Timescale is in millions of years before present. Node bars show 95% highest posterior distributions for major divergences. Reproduced with permission (Sanders et al., 2013). **B)** Relocation of V2R loci following re-assembly of *Hydrophis* major (Trim, 2025; the present author). **C)** Left: V2R gene loci over genomic position on chromosome 2. Right: ribbon plot visualising synteny at the beginning of chromosome 2. Note the region is significantly unsyntenic where V2Rs are present, even across *Hydrophis*. **D)** Hi-C chromatin-interaction heatmaps of chromosome 2 (top) and the Z sex chromosome (bottom). Gaps in colour, and dispersal from the midline indicate structural or repetitive complexity in the area. Note the difference in chromosome 2 compared to chromosome 7 & 8 (left and right of).



the largest mammalian V2R complements are found in rodents and the platypus, both of which exhibit expanded vomeronasal receptor repertoires consistent with strong reliance on dissolved-compound and pheromonal cues (Grus et al., 2007; Nei & Rooney, 2005; Yang et al., 2005). Against this broad vertebrate backdrop, squamates, and within them, *Hydrophis*, offer a sharp contrast for probing how V2R repertoire organisation maps onto ecological function, and major signal-medium shifts (air to water).

V2Rs in Squamata and Hydrophiinae

Squamate V2R data remains sparse, descending from method heterogeneity. VNO transcriptomic profiling of *Pantherophis guttatus* identified more than 116 distinct V2R transcripts, placing squamates among the most V2R-rich amniotes yet characterised (Brykczynska et al., 2013). However, transcriptomics is inherently time- and state-specific and systematically underestimates the full genomic complement. Whole-genome analysis of *Crotalus adamanteus* substantiated this, recovering approximately 687 putatively functional V2Rs (Hogan et al., 2025). These loci, organised in large tandem arrays are subject to widespread episodic positive selection in the ligand-binding domains, confirming that receptor specificity is diversifying under directional selection (Hogan et al., 2025).

Within the snake genome more broadly, V2Rs are localised to a small number of chromosomes (hydrophiines: to the Ch2 & ChZ sex chromosome) and assembled into dense tandem arrays, with *C. adamanteus* containing 64 loci in a ~4Mbp array on Ch2 (Hogan et al., 2021). Comparably in *Hydrophis*, V2R loci are concentrated into tandem arrays, rich in repetitive elements that have expanded primarily through tandem and segmental duplications (Li et al., 2021; Ludington et al., 2023). These arrays span several megabases and are interspersed with pseudogenes and truncated paralogues, making them particularly challenging to analyse. Within *Hydrophis*, published V2R counts derive from genome assemblies of variable quality and from bioinformatic pipelines that differ in their ability to resolve highly repetitive, tandem-array-nested loci. Older genome assemblies systematically fail to fully capture the repetitive nature of tandem arrays, underestimating the true copy number

variation (CNV). The magnitude of this discrepancy renders many early V2R-count estimates likely unreliable.

For example, Kishida et al. (2019) estimated just 137 V2R loci in sea snakes, fewer than the ~350 identified in their terrestrial relatives, implying marine-associated contraction of the vomeronasal receptor repertoire. Conversely, Li et al. (2021) reported more than 1,000 V2R copies in two *Hydrophis* species (*H. cyanocinctus*: 1,411; *H. curtus*: 1,065), denoting the opposite pattern: a massive expansion. More recently, Ludington & Sanders (2025, unpublished) identified 340 V2R copies on chromosome 2 alone in *H. major*, further complicating the picture, while Policarpo et al. (2024) concluded that marine tetrapod lineages exhibit convergent chemoreceptor-repertoire reduction when using a standardised prediction pipeline across more than 1,500 vertebrate genomes. However, Policarpo et al. (2024) applied relatively lenient criteria for genome assembly inclusion (e.g., >80% completeness), in addition to using *Hydrophis* assemblies that predate those of the other studies described.

These contradictions are not readily attributable to biological variation: a ten-fold spread in V2R counts within a genus that diversified within the last million years is structurally incompatible with genuine evolutionary divergence at this locus. The inconsistency instead likely reflects differences in genome quality stemming from the limitations of the bioinformatic tools used at the time. However, this discrepancy is not unexpected, given their complex structure and organisation of repeat- and gene-dense tandem arrays that are notoriously difficult to assemble accurately (Pan & Zhang, 2008; Rhie et al., 2021).

Structure & Mechanisms of Complex Gene Arrays

Repetitive elements constitute the dominant source of structural variation in vertebrate genomes, comprising 23–56% of squamate genomes (Peng et al., 2020), while conservative estimates denote

~25% of all gene duplication events in vertebrates are tied to tandemly arrayed genes (Pan and Zhang 2008). Transposable elements (TEs), are mobile repeats that reshape arrays through distinct molecular mechanisms that converge on copy-number instability. Long terminal repeat (LTR) retrotransposons, structurally analogous to retroviruses, carry internal promoters and transcription-factor binding sites capable of introducing novel *cis*-regulatory elements or duplicating flanking coding sequences (i.e., during transposition) (Chuong et al., 2016). In *Hydrophis*, LTRs constitute ~26–34% of the genome, with insertion age distributions placing the onset of expansion at ~12.5 Mya, coinciding with divergence from terrestrial Elapidae and bracketing marine transition (Peng et al., 2020; Ludington et al., 2023). When LTRs flank coding sequences, their sequence identity provides ectopic recombination substrates for non-allelic homologous recombination (NAHR), enabling tandem duplication of intervening genes (Dittwald et al., 2013; McGurk & Barbash, 2018; Mérot et al., 2020).

The structural convergence of TE-enriched V2R clusters across Squamata supports their role as genomic ‘*sandboxes*’ for chemosensory adaptation. Across squamates V2R and OR loci organise into tandem arrays identifiable as distinct chromosomal blocks (Hogan et al., 2021). Varanid lizards exhibit clustering on Ch2 and the ChZ/W sex chromosomes, where elevated TE density coincides with extensive copy-number variation and gene traffic between chromosomes (Matsubara et al., 2014; Zhu et al., 2025). Specifically, *V. komodoensis* (komodo dragon) ~13-gene V2R clusters on Ch2 coincide with high repeat density and elevated structural variation (Lind et al., 2019; Zhu et al., 2025), mirroring the ~4Mb array harbouring 64 V2Rs in *C. adamanetus* (Hogan et al., 2021), the “continuous cluster of 35 tandem copies” in *H. cyanocinctus* and the 77 in ChZ’s cluster in *H. curtus* (Li et al., 2021). These observations suggest that TE accumulation creates evolutionary hotspots where tandem redundancy simultaneously generates receptor diversity and substrates for NAHR.

Additionally, repetitive architecture influences local genomic stability. Structural variations identified in localised V2R clusters at the start of chromosome 2 in *H. major* include inversions spanning array-containing segments (Ludington & Sanders, 2025, unpublished). Such inversions suppress local recombination, preserving specific receptor haplotype combinations (similar to evolutionary strata

observed in varanid sex chromosomes), while translocations may relocate arrays into regulatory environments with distinct enhancer accessibility (Mérot et al., 2020). Consequently, TE-seeded arrays may exist in a dynamic context, whereby the same redundancy that generates functional diversity also creates genomic instability that can drive adaptive expansion.

Resolving Repeat-Dense Regions

The structural consequences of this repetitive architecture undermine genomic inference. Tandem V2R paralogs share 60–90% sequence identity, while intergenic TE insertions generate pairwise similarity between otherwise unrelated flanking regions (Alkan et al., 2011). When adjacent duplicates exceed 85–90% identity, assembly algorithms, particularly those utilising short-read data, treat overlapping reads from distinct copies as alternative paths through a single locus, collapsing them into one consensus sequence rather than resolving separate tandem members (Tørresen et al., 2019; Tremblay, 2010). This collapse is not merely a limitation of short-read technologies, even long-read assemblies require deliberate parameter optimisation to prevent premature joins in repeat-dense regions, with standard quality metrics such as N50 or BUSCO incapable of detecting these errors (Howe et al., 2021; Priyam et al., 2021; Tørresen et al., 2019). Priyam et al. (2021) demonstrate that modifying error-rate estimates in long-read assembler configurations can improve output quality >100% on some metrics, with optimal parameters differing between species, technologies, and repeat compositions. Thus, the same repetitive mechanisms that generate V2R diversity through tandem duplication simultaneously obscure that diversity from genomic study. For *Hydrophis*, where LTR bursts have created massive V2R arrays coincident with adaptation, modern technologies must be leveraged to better capture these regions and close the gap between assembly and ground-truth genome accuracy.

Genomic Resources: Importance & Availability

The cost of sequencing a eukaryotic genome fell approximately five orders of magnitude between 2001 and 2024, from roughly 100 million dollars per genome to under 1,000 dollars, outpacing even the rate predicted by Moore's Law (historical trend and a goal for the tech industry) (Muir et al., 2016; Wetterstrand 2022). This decrease, driven initially by the transition from Sanger to massively parallel short-read sequencing and then by the commercialisation of long-read platforms, facilitated the exponential growth in non-model species reference genomes. The Earth BioGenome Project (EBP), launched in 2018, formalised this trajectory as a ten-year initiative to sequence all 1.8 million eukaryotes, embodying an ambition framed around comprehensive representation rather than mere catalogue-completion (Lewin et al., 2018). Whilst parallel efforts, including the Vertebrate Genomes Project (VGP) and telomere-to-telomere (T2T) initiatives, concentrate on generating complete chromosome-scale assemblies, these efforts still reflect goals of quantity, rather than quality (Rhie et al., 2021).

Revisiting archived genomic resources offers an economic and ethical alternative to re-sequencing organisms, especially for rare or difficult-to-sample lineages (e.g., marine snakes). With sequencing costs continuing to decline and an increasing suite of sequence-agnostic quality-assessment tools, there may lie merit in re-processing existing data using novel assembly algorithms, chromatin-interaction data, and improved annotation pipelines, to yield superior genomes, without the need for new specimen collection.

Long read sequencing technologies have become essential for complex genomes (e.g., *Hydrophis*), as their reads span multi-kilobase-scale regions (i.e., tandem arrays), contrasting the fragmentive or collapsed products of assembly using short-reads (Rhie et al., 2021; Zhang et al., 2022). Long reads preserve haplotype phase and structural context enabling contiguous and accurate assembly (Zeng et al., 2024). When integrated with Hi-C chromatin-conformation data (maps the 3D spatial organization of chromatin, identifying physically proximal DNA regions), contigs can be oriented into scaffolds, resolving repeats and distinguishing true tandem duplication from misassemblies (Peart et al.,

2021; Zheng et al., 2024). Hi-C further improves the detection of structural variants and copy-number changes within repetitive regions, as spatial proximity patterns reveal breakpoints and rearrangements that are otherwise obscured by sequence similarity (Melo et al., 2020).

Assembly & Annotation: Technical Limits on Biological Inference

Errors introduced during genome assembly can propagate into downstream analysis and resulting conclusions, with this amplified in complex chromosomal regions. Three assembly artifacts (errors) dominate: collapses, in which multiple tandem copies or paralogs are merged (resulting in an underestimate of copy number); misjoins, where non-adjacent sequences are incorrectly connected (creating spurious sequences); and duplications of single-copy sequence to multiple contigs (resulting in an overestimate of CNV).

The expansion in genome availability is not, however, equivalent to an expansion in quality — a distinction rarely discussed. Across the reference genome landscape: a species is typically sequenced once, assembled with the tools available, submitted to a public database, and then used in extended downstream analyses while bioinformatics algorithms continue to advance independently. Rhie et al. (2021) noted, after analysing VGP reference assemblies, that they contained substantial errors in “unresolved complex repeats”, including false gene duplications, chromosomal rearrangements and missing sequence. Supporting this premise, Lawniczak et al. (2022) subsequently formalised EBP minimum standards, recognising that the majority of assemblies submitted did not satisfy modern scrutiny.

Similarly, Howe et al. (2021) demonstrate that even reference genome assemblies, which score highly by standard metrics, commonly harbour errors concentrated in repeat-rich regions, obscuring true biological signals from the data. For instance, the standard contiguity metric, ‘N50’, measures the shortest contig length such that all contigs longer than cover at least 50% of the total assembly length; although representative of contiguity, it is incapable of detecting errors or accuracy (Wang & Wang, 2023). Best practice bioinformatics now tends to follow the “3 C’s” as a criterion for assessing genome

assemblies, comprised of contiguity (size and number of contigs), completeness (presence of expected genes), and correctness (accuracy of the sequence) (Wang & Wang, 2023). Following these principles will generate superior reference genomes, facilitating more accurate downstream biological research.

Synthesis

Accurate reconstruction of complex gene families or regions requires long-read sequencing and chromatin-interaction data as the foundation to appropriate methodology. The two ensure inferences on copy number variation and functional organisation are biologically valid. The vomeronasal type-2 receptor gene family encodes chemosensory discrimination in sea snakes. Multiple convergent lines of evidence establish that this gene family was under strong selection during and after marine transition, however, the available assemblies are inadequate for accurately studying V2R evolution in *Hydrophis* snakes, as they do not offer sufficient V2R annotation, evident in the count discrepancy across *Hydrophis*. Current gene counts are incompatible with genuine biological variation within a genus that diversified within the last ~1 million years.

Research Proposal

Project Rationale & Overview

This project extends the premise that merit exists in revisiting older genomic data to produce new and updated reference genome assemblies. This project aims to produce six, annotated *Hydrophis* chromosome-scale reference genomes, and the production of the first standardised, quality-controlled, comparative V2R repertoire dataset for sea snakes. These improved resources will enable accurate comparative genomic and phylogenetic analyses to model V2R-family dynamics in *Hydrophis* sea snakes.

A targeted proof-of-concept undertaken by Trim (2025; the present author), reassembled *Hydrophis major*, and assessed chromosome 2. By optimising long-read integration and Hi-C scaffolding, Trim (2025) identified problematic regions, confirmed through misassignment (i.e., between *H.major* Ch2 & inter-species ChZ) colocalising with repetitive regions. Subsequent tool and parameter exploration, produced a new assembly with improvements across all quality metrics (Trim, 2025; present author). This resolution recovered 570 V2R loci on chromosome 2 alone, a 68% increase over the published assembly from identical input reads (Trim, 2025; present author). This establishes that assembly quality translates into inference accuracy, and aligns with the notion of systematic underestimation in *Hydrophis*'s V2R repertoire.

Research Aims

- Aim 1** Generate improved chromosome-scale genome assemblies and annotations for six *Hydrophis* species.
- Aim 2** Produce a high-confidence, standardised V2R gene catalogue across the full species panel using a consistent layered annotation framework.
- Aim 3** Model V2R gene family dynamics across the radiation and marine transition, and test structural genomic predictors of copy-number variation.
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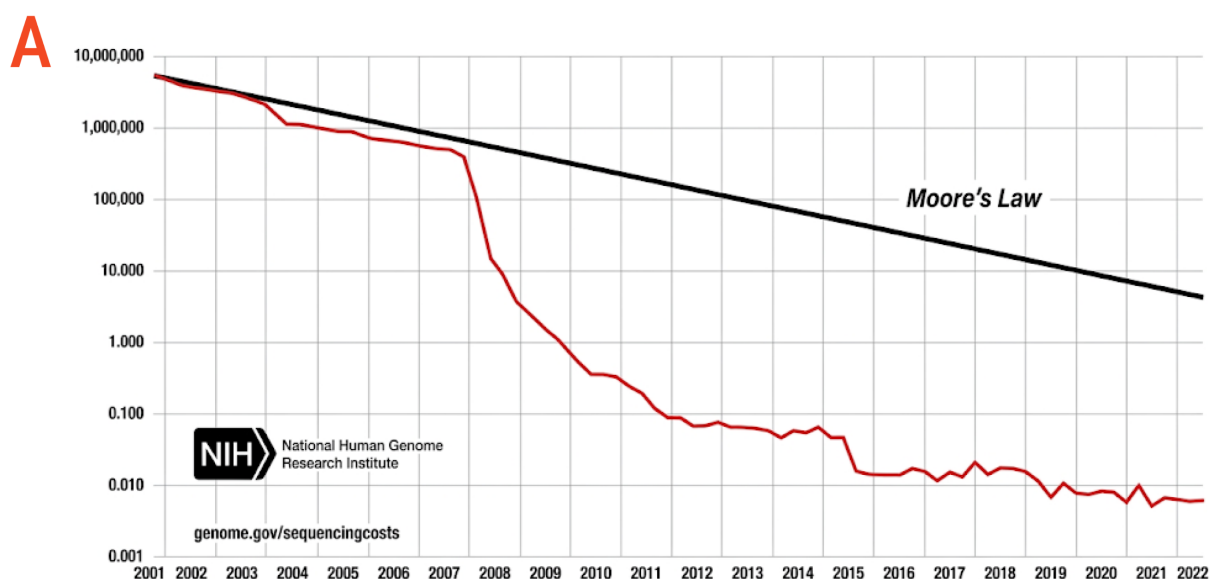
Project Hypotheses

- H1** Improved assemblies will reveal V2R repertoires larger than published — confirming that assembly artefacts systematically deflate existing counts, and hence V2Rs are underestimated in the literature.
- H2** *Hydrophis* species show elevated rates of V2R gene gain relative to terrestrial elapid outgroups and to neutral birth-death expectation, as an adaptive response to the marine environment.
- H3** Expanded V2R gene lineages in *Hydrophis* exhibit signatures of positive selection consistent with diversifying selection maintaining receptor specificity breadth across the radiation.
- H4** Local TE density in V2R-clusters is positively associated with V2R copy number across *Hydrophis* species, consistent with TE-mediating recombination, regulation and driving tandem array expansion.
-

Experimental Design

Data Acquisition

The species panel for the study comprises 5 *Hydrophis* lineages comprising three nominal species (*H. major*, *H. cyanocinctus*, and *H. ornatus*) and two lineages of *H. curtus* (Indian Ocean versus Australia-Pacific). For outgroups, *Aipysurus laevis* from the genus *Aipysurus*, a monophyletic clade of sea snakes in Hydrophiinae, and the terrestrial elapids *Pseudonaja textilis* (eastern brown snake) and *Notechis scutatus* (mainland tiger snake), will be used in the comparative genomics and phylogenetics section. The selection of species therefore resolves three contrasts: terrestrial versus marine, independent marine transitions, and within-*Hydrophis* radiation. Reads will be downloaded from the Sanders laboratory, and public datasets. RNA-seq libraries stored by the Sanders Laboratory will span



B

Technology	Mechanism	Primary Utility in Genome Assembly
Illumina	Short-read (150–300 bp), high-throughput sequencing.	High base-level accuracy (99.9%). Used for "polishing" long-read assemblies and correcting Indels.
ONT	Oxford Nanopore: Ultra-long reads (10 kb to 1 Mb+) via nanopore sensors.	Excellent for spanning highly repetitive regions and resolving structural variants. Raw accuracy is lower than HiFi but improving (Q20+).
PacBio CLR	Continuous Long Read: Single-pass SMRT sequencing.	Long reads (10–30 kb) with random error profiles. Largely replaced by HiFi; requires high coverage for consensus accuracy.
PacBio HiFi	High Fidelity: Circular Consensus Sequencing (CCS).	The "gold standard" for de novo assembly. Combines long-read lengths (15–20 kb) with 99.9% accuracy by reading the same molecule multiple times.
Hi-C	Chromatin-interaction mapping (proximity ligation).	Provides long-range spatial info. Essential for "scaffolding" genomic contigs into chromosome-scale sequences by identifying sequences physically close in the nucleus.

C

Species	Data Types	Source / Accession	Original Assembly
Raw Data Holdings (Sanders Lab & Public Repositories)			
<i>H. major</i>	PacBio HiFi, Hi-C	Bioplatforms: 102.100.100/351827, 351778; Hi-C: 351780	JAUPSR000000000
<i>H. curtus</i> (West)	ONT, Illumina, Hi-C	NCBI PRJNA780942	JAWKAS000000000
<i>H. ornatus</i>	ONT, Illumina, Hi-C	NCBI PRJNA780942	JAWKAR000000000
<i>H. curtus</i> (East)	ONT, Illumina, Hi-C	NCBI PRJNA573877, PRJNA616080, PRJNA608244, PRJNA622900	JABAHG000000000
<i>H. cyanocinctus</i>	PacBio CLR, Illumina, Hi-C	NCBI PRJNA573877, PRJNA616080, PRJNA608244, PRJNA622900	JAAZTL000000000
<i>H. elegans</i>	Illumina, ?Long-read/Hi-C	NCBI PRJNA984433	JAUPSR000000000
Outgroups (Public Assemblies)			
<i>Aipysurus laevis</i>	Chromosome-scale	NCBI GCA_040207615.1	—
<i>Pseudonaja textilis</i>	Elapid reference	NCBI GCA_049901715.1	—
<i>Notechis scutatus</i>	Chromosome-scale	NCBI GCA_049901735.1	—

Figure 3. A) Cost of sequencing per megabase, compared to Moore's Law. Figure adapted from Wetterstand, 2022. **B)** Glossary of genomic sequencing technologies, their mechanisms, and their functionality in genome assembly. **C)** Table of data availability for the research project. Accession numbers and data bases where raw sequencing data will be downloaded from.

vomeronasal organ, retina and venom gland tissue from multiple hydrophiines and other snake species; following the principle that RNA evidence markedly improves splice-site inference during gene annotation (Zimin et al., 2025). Repeat libraries provide consensus sequences, reducing the noted false-positive rate when using generic, or entirely *de novo* constructed libraries with squamate genomes (Bao et al., 2015; Ludington et al., 2023). The final genome assemblies and annotation files will be made publicly accessible and uploaded to NCBI following completion of the project.

Methods

Genome Assembly

Genomes will be assembled using an automated, containerised, and modular pipeline, written in NextFlow DSL2 (Di Tommaso et al., 2017), for reproducibility across high-performance computing (HPC) environments. Modularity allows each process to be swapped or reparameterised without disturbing upstream stages, and containerisation ensures reproducibility across environments. The pipeline extends on the framework validated in the 2025 proof-of-concept (POC) re-assembly of *H. major* (Trim 2025; present author).

De novo contig assembly: Read type determination of assembler:

- ❖ HiFi: Hifiasm (Cheng et al., 2021), with parameters selection established from POC results signifying fragmentation-permissive settings yield a superior assembly by preventing premature contig joins in repeat-dense regions, a finding consistent with Priyam et al. (2021) (Trim, 2025; present author).
- ❖ ONT or CLR: NextDenovo (Hu et al., 2024). *Note: The de novo assembly of H. ornatus will be computed using Flye (Kolmogorov et al., 2019) as opposed to NextDenovo, to explore any potential effect the assembler has on quality.*

Contig correction:

Inspector (Chen et al., 2021) will be used to scan for read-inconsistent contigs and correct misassemblies. CRAQ will then be used to break the Inspector corrected output at positions where coverage depth and clipping (frequency of read-assembly disagreement at contig ends) together indicate structural errors (Li et al., 2023). NextPolish (Hu et al., 2020) will then polish the final *de novo* assembly using raw long reads, to correct indels prior to scaffolding. Polishing order is deliberate: base-level correction precedes scaffolding because Hi-C scaffold-breaking tools assume the input sequence is locally correct (Rhie et al., 2021).

Hi-C scaffolding:

HapHiC (Zeng et al., 2024) joins contigs into scaffolds, or pseudo-chromosomes, using Hi-C contact information (measures the physical proximity of DNA segments by recording how frequently different parts of the genome touch or "interact" with one another in 3D space). Scaffold-level errors (e.g., incorrect orientations, inversions, misjoins, etc.) are corrected using the 'MoreHiC' module from AutoHiC, which leverages deep-learning on images of Hi-C contact maps, achieving >90% accuracy of correction (Jiang et al., 2024).

Manual curation:

PretextView then allows for manual inspection, breaking, inversion and orientation of scaffolds using Hi-C contact heatmaps. Howe et al. (2021) demonstrated that curation recovers on average 6% additional contig length and corrects a median of 4 misjoins per vertebrate assembly; whilst invisible to QC metrics (e.g., N50 & BUSCO) these errors impact syntenic inference around tandem gene arrays. TGScapcloser then fills residual scaffold gaps with long-read support, and a final round of NextPolish, using short-reads, amends small-scale base errors (Xu et al., 2020).

Quality control:

QC metrics are computed iteratively after every stage, ensuring gradual improvement in assembly quality. The QC metrics selected support the "3C" framework: contiguity, completeness, correctness. Contiguity: total size, total contigs/scaffolds, N50 and auN (the area under the N-curve, which plots sequence lengths from longest to shortest against the cumulative percentage of the total assembly, providing a weighted average of contiguity). (QUAST; Gurevich et al., 2013). Completeness: BUSCO (presence of single-copy ortholog genes versus expected) via Compleasm (Huang et al., 2023) against the squamata_odb12 lineage set, plus k-mer completeness and QV, which demonstrate how representative the underlying assembly is of the sequence data (MerquryFK against FastK fastk-mer databases (Myers 2020). Correctness: MerquryFK QV and Mosdepth for coverage statistics (e.g., breadth of coverage; Pedersen, 2018), evaluate if more reads map (and map without e.g. clipping/multi-mapping) to the assembly and how uniform coverage is across the genome.

Genome Annotation

Repeat masking:

RepeatModeler2 (Flynn et al., 2020) has previously been used to perform *ab initio* repeat identification (Bao et al., 2015; Galbraith et al., 2020; Ludington et al., 2023). RepeatMasker (Chen et al., 2004) then soft-masks (substituting for lowercase letters (e.g., atgc) to represent repetitive regions) the assembly before gene prediction.

Gene annotation:

EviAnn performs RNA-evidence-driven gene annotation using mapped RNA-seq libraries and a curated protein database of V2Rs (Zimin et al., 2025). Eviann was selected due to its speed and accuracy, over standard alternatives (MAKER or BRAKER2) (Zimin et al., 2025).

Genome Assembly Assessment

Old-versus-new assembly comparison:

Each reassembled genome is compared to its published predecessor on QC metrics as outlined previously in the “Genome Assembly” section, as well as V2R count, and on chromosome-scale synteny (Figure 2(b)). Whole genome alignment (with FastGA; Myers et al., 2025), and ribbon plotting syntenic improvement will allow visualisation of structural rearrangements (Goel et al., 2019).

New-assembly Comparative Genomics:

Across the new panel, QC metrics, similarly will be used to broadly compare genomes across species, with particular attention to V2R-carrying regions on chromosomes Ch2 and ChZ. Synteny analysis will expose structural similarities and variation across the species panel, which poses particular use for *Hydrophis*, as no species have been karyotyped yet (i.e., the actual chromosome count is unknown).

V2r Gene-Family Dynamics & Phylogenetics

V2R quantification & cross-validation:

Candidate V2R loci identified by EviAnn will be subjected to cross-validation to reduce common annotation errors (e.g., misclassification of V1Rs as V2Rs, fragmented/merged models, or pseudogenised V2R loci). Predicted gene models will have their protein sequences searched against the UniProt curated database (The UniProt Consortium [UniProt], 2025) using BLASTP (REF) to identify putative V2R genes. For this subset of genes, 7-transmembrane domains (7TMs; see figure 1) will be annotated in each sequence using HMMER (REF) and the Pfam database (REF), and screened to ensure they have exactly one 7TM domain (Gertz et al., 2006). A profile hidden Markov model (pHMM), constructed from the consensus multiple sequence alignment (MSA) of curated vertebrate V2Rs (HMMER; Potter et al., 2018), will then re-assess all candidate loci. This combined approach will mark and correct problematic genes, producing an accurate V2R repertoire per species that explicitly classifies intact and artifacts for robust evolutionary analysis.

V2R-subfamily & Orthology Grouping:

As per the method used by Li et al. (2021), grouping together extracted V2Rs, aligning them (MSA) and building an unrooted phylogenetic tree, will allow visualisation of domain clustering of the protein, and potential functional implications. OrthoFinder will be utilised to cluster genes into orthogroups across marine and terrestrial hydrophiines (Emms & Kelly, 2019). These orthogroups will then be passed to CAFE (Mendes et al., 2021), which models variation in evolutionary rates *among* gene families, and fits a birth-death model along the dated species tree to estimate per-lineage gain and loss rates for each gene family, including V2Rs. Sanders laboratory's species tree, or a new one – computed with Read2Tree (Dylus et al., 2023) – will be utilised for this. For each orthogroup, MSA of the protein sequence will then be translated back to the coding sequence using PAL2NAL (Suyama et al., 2006) to produce a codon alignment.

Transposable element–CNV association:

A one-sided permutation test will be used to establish if transposable elements are enriched in the vicinity of V2R genes on Chr2 compared to the genomic background. The observed number of TEs intersecting the region of interest (ROI; genomic region housing the V2R array) in each species will be calculated using BEDTools (Quinlan & Hall, 2010). An empirical null distribution will then be estimated by repeatedly shuffling a region of equal length to the ROI along the chromosome randomly using *bedtools shuffle*, calculating the number of TE overlaps in each permutation. Significance will be assessed with a one-sided Monte Carlo permutation p-value, whereby we count the number of test statistics as or more extreme than our initial observed statistic, divided by the total number of permutations. Fold enrichment will be calculated by dividing the observed statistic by the mean of the null distribution.

Selection analysis:

PAML (phylogenetic analysis by maximum likelihood, Yang, 2007) tests for episodic positive selection on V2R branches in the marine lineages and individual codon sites under positive or purifying

selection. For each aligned and curated V2R Orthogroup, branch and branch-site models will be used to test for positive selection.

Synteny and structural context:

Chromosomal synteny in V2R dense regions will be used to understand if there is a role of genomic architecture on V2R gene family dynamics between species. Whole-genome alignments between species will be passed to SyRI to call structural variants and ribbon plotting synteny, before overlaying orthogroup information from the CAFE output (e.g., gene gain/loss events) (Goel et al., 2019).

This will place V2R gain/loss events, and their orthogroups in their chromosomal context. This will involve alignment (with FastGA; Myers et al., 2025), SyRI to call structural variation, and ribbon plotting synteny, including overlaying orthogroup information from the CAFE output (e.g., gene gain/loss events) (Goel et al., 2019).

Timeline Contingencies: If the NextFlow pipeline execution becomes computationally intensive or resource-limited, individual species assemblies will use modular scripts with identical tool/parameters, prioritising pipeline reproducibility over full automation.

Research Significance & Outcomes

This project operates at the intersection of two independent problems, understanding the evolution of secondarily marine chemosensory systems, and the reliability of non-model genomic resources, with significance extending in both directions. At the most immediate level, it corrects a foundational question: how V2Rs evolved, across the *Hydrophis* radiation. This is currently answered contradictorily by the only studies that have attempted to quantify it.

The broader significance lies in the model system itself. *Hydrophis* presents an unusually tractable set of conditions for studying gene family adaptation: the phylogeny is resolved, the ecological shift is discrete, the selective pressure is specific, and the radiation occurs in a compressed temporal window.

The findings from this project are relevant to any vertebrate lineage undergoing major sensory shift, or with structurally complex genomes.

Beyond resolving a contentious debate in sensory evolution, the products of the project workflow will serve as a scalable blueprint for resolving complex loci in non-model organisms. Furthermore, by generating publicly-available, reference genomes for vulnerable Indo-Pacific marine species, this project provides vital infrastructure for future studies. *Hydrophis* include some of the least taxonomically characterised squamates, and improved reference genomes are common prerequisites for population genomic studies underpinning conservation assessments.

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